

Firms as Foragers

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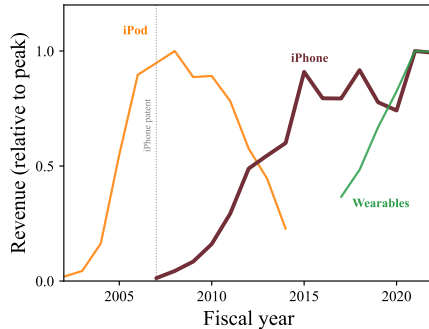
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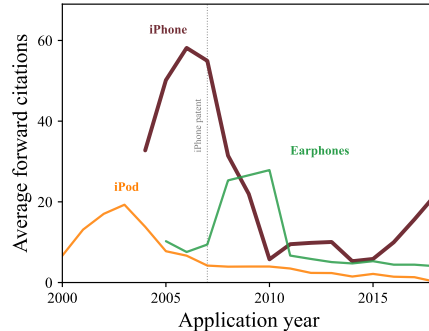
Apple's history of invention

► Movements in space

"If you do something and it turns out pretty good, then you should go do something else wonderful, not dwell on it for too long. Just figure out what's next."
(Steve Jobs)



(a) Hardware revenue, by product



(b) Average citations, by patent cluster

Motivation

- **Organizational literature** stresses firms' continual choice of **exploiting** a local stock of knowledge until it thins vs. **exploring** for new ideas *[e.g. March, 1991]*
 - behavioral ecology: theory of animals optimally foraging for food ▶ "Foraging"
- Many **growth** questions hinge on this decision of when a firm should move on
 - How much of growth comes from discovering new territory vs. deepening old ground?
 - Has productivity growth slowed because ideas are locally harder to find *[Bloom et al., 2020]* or because exploration has become more burdensome *[Jones, 2009]*?
 - How does growth change if AI eases exploration or, instead, exploitation?
- But the explore-exploit cycle has no counterpart in most **macro models of growth**

An empirically grounded framework where growth arises from firms foraging in idea space.

- **Develop tractable theory of growth centered on the **explore**–**exploit** cycle**
 - tractable optimal stopping problem: when to leave as potential of patch diminishes
 - embedded in standard GE model with expanding, quality-differentiated varieties
- **Document new empirical facts about firms' innovation behavior**
 - use text of patents to construct an empirical representation of idea patches
 - track firms' movements and innovation outcomes within and across patches
- **Apply theory & data to quantitatively study the *pace* and *modes* of growth**

- ① **Systematic evidence:** firms face **diminishing returns to innovation** *within* a patch
- ② **Vintage composition of growth:** **patenting in new patches is key to growth:** at a 20-year horizon, it accounts for over half of growth from quality improvements
- ③ The historical **productivity slowdown** is better explained by **exploration getting harder** than by **worsening exploitation**

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The potential for quality improvement gets exhausted locally, but firms continually entering new territory sustains growth.

- Start with **theory**: individual firm's foraging problem
- Then turn to **empirics** of foraging – through lens of theory
- Embed individual firm's problem into **GE** model of aggregate growth (briefly today)
- **Calibrate** the model and **apply** theory & data to derive macro implications
 - origins of the productivity slowdown
 - vintage composition of growth

Theory: Firm Foraging

Theory: environment

- Consider the problem of a single firm—GE, BGP, etc. come later—in continuous time
- A **firm** is a collection of products indexed by quality q_j
 - at most one **active product** whose quality the firm tries to improve through R&D
 - legacy portfolio of products that are sold but no longer improved
 - products may be lost (δ) or firm dies (δ_F)
- **Product** flow profit $\pi(q)$ is increasing in its quality
- A firm inhabits a space of sparsely distributed **idea patches**, each has a “density” X
 - arbitrarily many patches, no re-encounter
 - initial densities are heterogeneous: $x_{0,k}$
- **Composite-search strategy:** firm either **exploits** a patch or **explores** for a new one
→ the only decision is *when* to optimally switch to max. PDV of profits

Environment: foraging technology

- **Exploitation:** improve the quality of active product
 - patents arrive at Poisson rate γ
 - each improves the active product with a multiplicative step size *increasing* in X

$$q \rightarrow \theta(X) q, \quad \theta(X) > 1, \quad \theta'(X) > 0$$

- patch gets depleted at rate $\mu < 0$:

$$dX(t) = \mu dt + \underbrace{\sigma_x dW_x(t)}_{=0 \text{ today}} \quad \text{with} \quad X(0) = x_{0,k}$$

- **Exploration:** search for a new patch
 - at rate λ the firm finds a new patch with *randomly drawn* initial density $x_{0,k}$
 - abandoned line becomes legacy at quality q_{exit}
 - new product inherits quality $q_{\text{new}} = \underbrace{\Lambda}_{=1 \text{ today}} q_{\text{exit}}$ ($\Lambda < 1$: limited capability transfer)

Value functions in normalized units

- Too many state variables? Patch density X , quality q , calendar time t ?
- **Demand:** CES with elasticity σ + monopoly pricing $\rightarrow$ flow profits π proportional to economic quality $\tilde{q} \equiv q^{\sigma-1}$:

$$\pi(q) = \kappa q^{\sigma-1} \equiv \underbrace{\kappa}_{\text{agg. profit shifter}} \underbrace{\tilde{q}}_{\text{quality in profit units}}$$

- **R&D costs:** R&D costs scale linearly in $\tilde{q}$ [cf. *Klette-Kortum, 2004*], with flow costs $\hat{c}_I \kappa \tilde{q}$ (exploitation), $\hat{c}_E \kappa \tilde{q}$ (exploration)
- **Normalized values** measure values in units of the line's own flow profit, e.g.

$$\hat{W}(X) \equiv \frac{\overbrace{\mathcal{A}(\tilde{q}, X, t)}^{\text{value of active line}}}{\kappa(t) \tilde{q}}$$

Patch density X is the only state $\rightarrow$ one-dimensional stopping problem.

Value functions

- **Legacy product value** $\hat{W}_0$: pure annuity that pays flow 1 (normalized profit)

$$\hat{W}_0 = \frac{1}{\hat{\rho} + \delta + \delta_F}$$

- normalized values are discounted at $\hat{\rho} \equiv \rho + (\sigma - 1)g$

- **Exploitation value** $\hat{W}(X)$

$$\underbrace{(\hat{\rho} + \delta + \delta_F) \hat{W}(X)}_{\text{required return}} = \underbrace{(1 - \hat{c}_I)}_{\text{profit net of R\&D}} + \underbrace{\gamma [\theta(X)^{\sigma-1} - 1] \hat{W}(X)}_{\text{innovation gains: step rises with density}} + \underbrace{\delta \hat{W}^E}_{\text{if destroyed, explore}} + \underbrace{\mu \hat{W}'(X)}_{\text{depletion drift } (\mu < 0)}$$

- **Exploration value** $\hat{W}^E$

$$(\hat{\rho} + \lambda + \delta_F) \hat{W}^E = \underbrace{-\hat{c}_E}_{\text{search cost}} + \underbrace{\lambda}_{\text{finding rate}} \underbrace{\sum_k p_k \hat{W}(x_{0,k})}_{\text{expected prize}}$$

Optimal stopping

- **Optimal policy:** abandonment value $\hat{W}_0 + \hat{w}^E$ is independent of X , so the policy is a threshold x^* : exploit while $X > x^*$ [e.g. Stokey, 2008]
- **Satisfies** value matching and threshold FOC:

$$\hat{W}(x^*) = \hat{W}_0 + \hat{w}^E$$

$$\hat{W}'(x^*) = 0$$

⇒ **Leave when the marginal delay in switching yields zero net benefit**

- $\sigma_x > 0$: value matching, smooth pasting $\hat{W}'(x^*) = 0$, boundedness condition at ∞

- Generalizes the **Marginal Value Theorem** [Charnov, 1976], with the outside option an equilibrium object rather than an exogenous intake rate

Firm-level growth rate

What is the avg. rate at which a firm accumulates log quality over an “innovation cycle”?

$$g_q \equiv \frac{E[G]}{E[T]} = \frac{\text{expected log-quality gain over a cycle}}{\text{expected duration of consecutive exploitation and exploration spells}}$$

With $\sigma_x = 0$, spell length is deterministic $T(x) = \frac{x-x^*}{|\mu|}$, so

$$\underbrace{m(x) = \int_0^{T(x)} e^{-(\delta+\delta_F)t} dt}_{\text{expected exploitation spell duration}}, \quad \underbrace{h(x) = \int_0^{T(x)} e^{-(\delta+\delta_F)t} \gamma \log \theta(x + \mu t) dt}_{\text{expected log-quality gain}}$$

and averages over patch types: $\bar{\tau}^I = \sum_k p_k m(x_{0,k})$ and $\bar{h} = \sum_k p_k h(x_{0,k})$

$\Rightarrow$

$$E[G] = \bar{h}, \quad E[T] = \bar{\tau}^I + \frac{1 - \delta_F \bar{\tau}^I}{\lambda + \delta_F}$$

Analytical characterization: assumptions

- We'd like to understand how primitives affect optimal leaving and growth
- We can obtain **closed-form characterizations...**
- ...if we also assume that...
 - deterministic drift: $\sigma_x = 0$ *[already in slides]*
 - no capability loss upon switching: $\Lambda = 1$ *[already in slides]*
 - exponential step size function $\theta(X) = e^{\phi X} \rightarrow \eta \equiv (\sigma - 1)\phi$
 - homogeneous patches: initial density x_0 , $0 < x^* < x_0$
- ...and adopt a myopic approximation: innovation rents $\hat{\Omega} \equiv \hat{W} - \hat{W}_0 \ll \hat{W}_0$
 - the firm treats $\hat{W}$ as being approximately equal to the passive legacy value $\hat{W}_0$
 - verified comparative statics numerically

Analytical characterization: stopping threshold

Proposition: Optimal stopping threshold

Under the myopic approximation:

$$x^* = \frac{1}{\eta} \ln\left(1 + \frac{R_{\text{eff}}}{\gamma}\right), \quad R_{\text{eff}} = \underbrace{\frac{(\hat{\rho} + \delta_F)\lambda}{\hat{\rho} + \lambda + \delta_F}}_{\text{exploration option}} + \underbrace{(\hat{\rho} + \delta + \delta_F)\hat{c}_I}_{\text{exploitation cost}} - \underbrace{\frac{(\hat{\rho} + \delta_F)(\hat{\rho} + \delta + \delta_F)\hat{c}_E}{\hat{\rho} + \lambda + \delta_F}}_{\text{exploration search cost}}.$$

⇒ **Better exploitation technology** – e.g. more frequent patents ($\gamma \uparrow$) or steeper payoff profile ($\phi \uparrow$) – raises payoff to staying at every $X \rightarrow$ **lower x^* , firm stays longer**

⇒ **Faster exploration (λ)** improves the effective outside option $R_{\text{eff}} \rightarrow$ **higher x^* , firm abandons patch sooner**

Proposition: Firm-level growth rate

Under the myopic approximation and setting $\delta_{\text{exit}} = 0$:

$$g_q = \underbrace{\gamma\phi \frac{x_0 + x^*}{2}}_{g_{\text{exploit}}: \text{ within-patch productivity}} \times \frac{\lambda(x_0 - x^*)}{\underbrace{\lambda(x_0 - x^*) + |\mu|}_{f_I: \text{ exploitation time share}}}$$

⇒ **Better exploitation technology** ($\gamma, \phi \uparrow$) lowers x^* and raises g_{exploit} → **growth rises** as exploitation spells *lengthen*

⇒ **Faster exploration** ($\lambda \uparrow$) raises x^* → **growth rises** as exploitation spells *shorten*

Empirical facts

- We need an empirical representation of the space of ideas over which firms search
→ We integrate patent-text data, processed with NLP tools, and firm-level data
- **Data:** [▶ Details](#)
 - PatentsView: patent metadata, citation links, CPC classifications
 - proxies for patent *novelty*, e.g., creativity *[Kalyani, 2024]* and *quality*, e.g. 5-year forward citations, patent value *[Kogan et al., 2017]*, breakthrough importance *[Kelly et al., 2021]*
 - CRSP/Compustat database: firm-year financials
- Construct a representation of idea space by applying NLP to patent text
 - 1 embeddings to turn patent text (title + abstract) into vectors
 - 2 clustering of these (continuous) vectors → empirical analogue to (discrete) patches

Apple: the iPhone multitouch patent and its text



US007479949B2

(12) United States Patent Jobs et al.

(10) Patent No.: **US 7,479,949 B2**
(45) Date of Patent: ***Jan. 20, 2009**

(54) **TOUCH SCREEN DEVICE, METHOD, AND GRAPHICAL USER INTERFACE FOR DETERMINING COMMANDS BY APPLYING HEURISTICS**

(75) **Inventors** Steven P. Jobs, Palo Alto, CA (US); Scott Forstall, Mountain View, CA (US); Greg Christie, San Jose, CA (US); Stephen O. Lemay, San Francisco, CA (US); Scott Herz, San Jose, CA (US); Marcel van Os, San Francisco, CA (US); Bas Ording, San Francisco, CA (US); Gregory Novick, Santa Clara, CA (US); Wayne C. Westerman, San Francisco, CA (US); Inman Chaudhri, San Francisco, CA (US); Patrick Lee Coffman, Menlo Park, CA (US); Kenneth Kocienda, Sunnyvale, CA (US); Nitin K. Ganatra, San Jose, CA (US); Freddy Allen Anzures, San Francisco, CA (US); Jeremy A. Wyld, San Jose, CA (US); Jeffrey Bush, San Jose, CA (US); Michael Matas, San Francisco, CA (US); Paul D. Marcos, Los Altos, CA (US); Charles J. Pisula, San Jose, CA (US); Virgil Scott King, Mountain View, CA (US); Chris Blumenberg, San Francisco, CA (US); Francisco Ryan Tolmasey, Cupertino, CA (US); Richard Williamson, Los Gatos, CA (US); Andre M. J. Boule, Sunnyvale, CA (US); Henri C. Lamiraux, San Carlos, CA (US)

(73) **Assignee** Apple Inc., Cupertino, CA (US)

(65) Prior Publication Data

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(63) Continuation of application No. 11/850,635, filed on Sep. 5, 2007.

(60) Provisional application No. 60/937,993, filed on Jun. 29, 2007, provisional application No. 60/937,991, filed on Jun. 29, 2007, provisional application No. 60/879,469, filed on Jan. 8, 2007, provisional application No. 60/879,253, filed on Jan. 7, 2007, provisional application No. 60/824,769, filed on Sep. 6, 2006.

(51) **Int. Cl.**
G09G 5/00 (2006.01)
G06F 3/048 (2006.01)

(52) **U.S. Cl.** 345/173; 345/169; 715/786; 715/784

(58) **Field of Classification Search** 345/156, 345/157, 173-181
See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

5,528,260 A *	6/1996	Kent	345/684
5,655,094 A *	8/1997	Cline et al.	715/786
5,805,161 A *	9/1998	Tiphane	715/786
6,278,443 B1	8/2001	Amro et al.	345/173
6,466,203 B2	10/2002	Van Fe	345/173
6,559,869 B1 *	5/2003	Lui et al.	715/785
6,597,345 B2 *	7/2003	Hirschberg	345/168
6,657,615 B2	12/2003	Harada	345/173

(57)

ABSTRACT

A computer-implemented method for use in conjunction with a computing device with a touch screen display comprises: detecting one or more finger contacts with the touch screen display, applying one or more heuristics to the one or more finger contacts to determine a command for the device, and processing the command. The one or more heuristics comprise: a heuristic for determining that the one or more finger contacts correspond to a one-dimensional vertical screen scrolling command, a heuristic for determining that the one or more finger contacts correspond to a two-dimensional screen translation command, and a heuristic for determining that the one or more finger contacts correspond to a command to transition from displaying a respective item in a set of items to displaying a next item in the set of items.

20 Claims, 293 Drawing Sheets

An embedding model turns the patent text (title+abstract) into a high-dimensional vector.

(57)

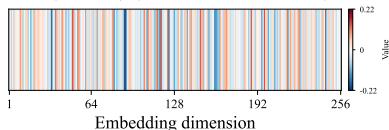
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$v(\text{US } 7,479,949) = (-0.074, -0.006, 0.061, -0.013, \dots, -0.097)$



From patent text to a point in idea space

Applied to all patents, this places related ones as neighbors in idea space.

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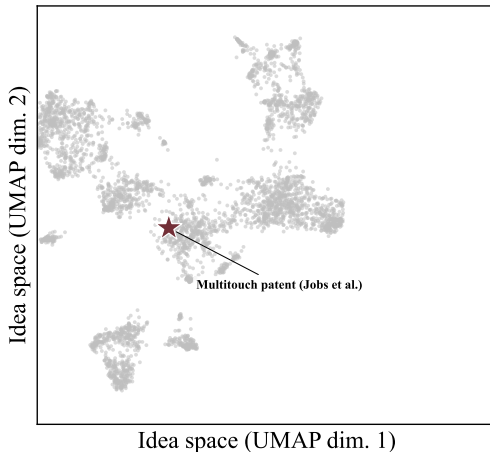
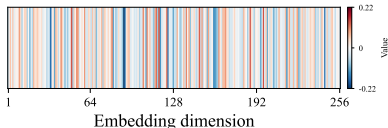
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From patent text to a point in idea space

Clustering then partitions the space into patches.

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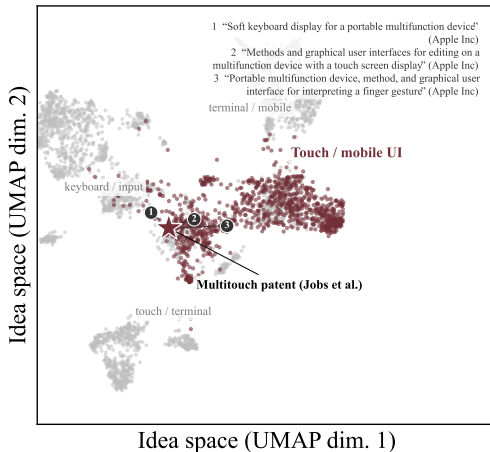
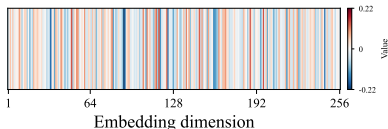
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With idea patches at scale, we study 3 questions

We can retrace each firm's trajectory through the idea space and relate it to innovation outcomes.

- ① **Within-patch dimension of foraging:** how do the characteristics of a firm's innovations evolve as it exploits a given idea patch?
- ② **Dynamics of exploration across patches:** how often do firms enter new patches, and how long do they stay?
- ③ **Firm-level outcomes:** how does foraging for ideas show up in annual firm panel data?

Fact 1 concerns the within-patch dimension of foraging

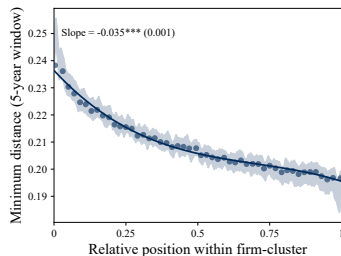
How do the characteristics of a firm's innovations evolve as it exploits a given idea patch?

→ For each firm and cluster, order patents by application date, assign a relative position & relate outcomes to that position (with application-year and firm-by-cluster FEs)

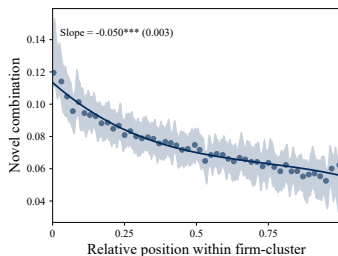
Fact 1: Diminishing returns to exploitation at the patch level

[► Patch definitions](#)[► CPC sections](#)

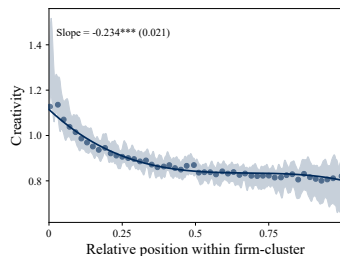
Patents decline in *novelty* as the firm advances through a patch.



(a) Min. distance



(b) Novel combinations



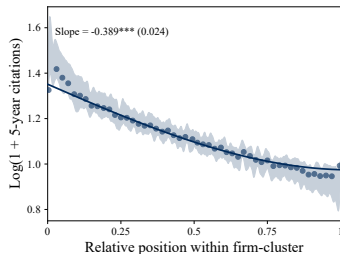
(c) Creativity

Fact 1: Diminishing returns to exploitation at the patch level

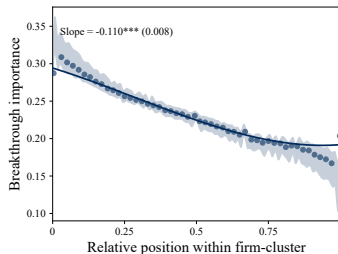
► No declining effort

► Selection

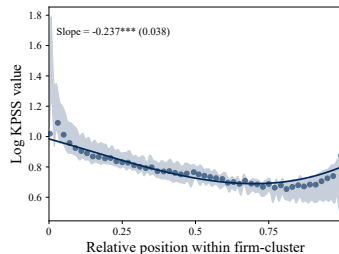
Patents decline in *quality* as the firm advances through a patch.



(a) Citations



(b) Breakthrough importance



(c) Market value (KPSS)

Fact 2 concerns the dynamics of exploration across patches

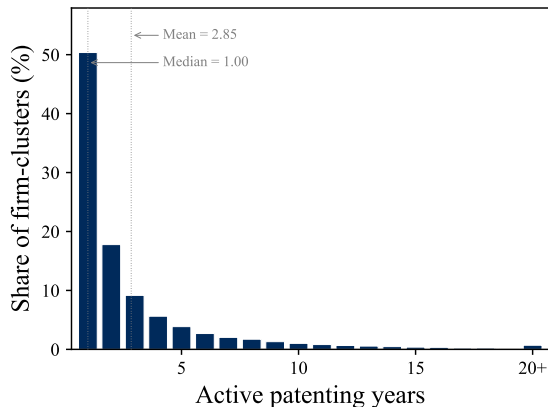
How long do firms stay in patches?

→ We measure *entry* and *exit* into patches – in the baseline, simply by the first/last year of firm j patenting in cluster c – and track the number of years with at least one patent between entry and exit: “active years”

Fact 2: Most patch spells are short; a minority run long

► Cluster quality and duration

Many cluster spells are short, but a minority develop into extended exploitation.



How does foraging for ideas show up in annual firm panel data?

→ Aggregate patent- and cluster-measures to the firm-year level and trace the dynamic association between new-cluster entry and subsequent firm-year outcomes using local projections:

$$\Delta_h Y_{jt}^m = \alpha_j^m + \delta_t^m + \beta_h^m \text{EntryRate}_{jt} + X'_{j,t-1} \Gamma_h^m + \varepsilon_{jt}^{m,h},$$

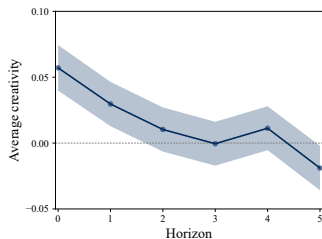
where

$$\text{EntryRate}_{jt} = \frac{\sum_c \text{Entry}_{jct}}{\text{ActiveClusters}_{j,t-1}}, \quad \text{ActiveClusters}_{jt} = \sum_c \mathbf{1}\{N_{jct} > 0\}.$$

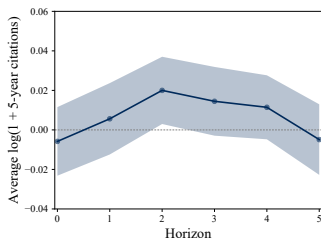
Fact 3: Entering new patches replenishes firm-level innovation

► Other outcomes

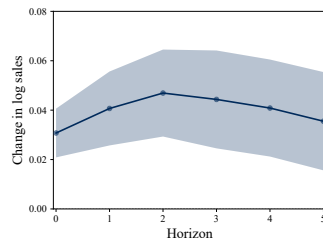
Following entry into new patches, patenting quality rises; sales follow with a lag.



(a) Creativity



(b) Citations



(c) Sales

Recap & interpretation of empirical facts

- ① **Within-patch dimension of foraging:** how do the characteristics of a firm's innovations evolve as it exploits a given idea patch?
→ **firms face diminishing returns to exploiting any given patch** ($\mu < 0$)
- ② **Dynamics of exploration across patches:** how often do firms enter new patches, and how long do they stay?
→ **firms move to new patches regularly;** often stay briefly, occasionally longer
- ③ **Firm-level outcomes:** how does foraging for ideas show up in annual firm data?
→ on average, firms are rewarded for exploration

Theory: BGP

General-equilibrium environment

How do many firms' foraging cycles aggregate to economic growth?

- Casting the explore-exploit cycle as a 1-dimensional optimal stopping problem → sufficiently **tractable** to embed into a standard GE endogenous growth environment
 - firm took as given g (via $\hat{\rho} = \rho + (\sigma - 1)g$) and labor market (via $\hat{c}_I = c_I/L^P$, $\hat{c}_E = c_E/L^P$)
- Representative household (log utility, discount rate ρ , inelastic labor supply L)
- Final good CES over varieties, each produced by monopolist with $y(j) = \ell(j)$
- **Aggregate quality index** and implied aggregates

$$Q \equiv \left(\int_0^{\mathcal{N}} q_j^{\sigma-1} dj \right)^{1/(\sigma-1)}, \quad Y = Q L^P, \quad w = \frac{\sigma-1}{\sigma} Q$$

- Endogenous firm mass:
 - free entry at sunk cost K_R
 - entrants start at $q_0 = \bar{\lambda}_E \bar{Q}_t$ where $\bar{Q}_t = \mathcal{N}_t^{-1/(\sigma-1)} Q_t$ is average product quality
 - exit via δ_F

Balanced-growth path equilibrium (intuition)

We consider a **balanced growth path (BGP) equilibrium** on which the aggregate quality index Q , and hence consumption, grow at a common rate g .

- Firm block delivers policies given (g, L^P)
- Stationary quality distribution delivers g
- Labor market clearing delivers L^P
- Free entry pins F

[► Details](#)[► Details](#)[► Details](#)

Growth rate characterization carries over

The characterization of the firm-level growth rate carries over up to interpretable wedges.

Proposition: Aggregate growth rate and g_q

In an economy with (i) $\sigma \rightarrow 1$ (the CES index reduces to a geometric mean), (ii) $\delta_F = 0$ (no firm death), and (iii) no legacy products (the quality index only contains varieties from active patches), the aggregate growth rate g is equal to the per-firm renewal-reward rate g_q .

Wedges:

- 1 For $\sigma > 1$, growth rate of CES power mean picks up x-sectional variance of log quality
- 2 With $\delta_F > 0$, death-and-replacement-during-exploration leads to quality dilution
- 3 Economy-wide average index includes legacy products

Quantitative applications

Calibration: overview of approach

The model's parameters can be disciplined using the empirically constructed “foraging moments” along with standard moments.

- Simplified model: $\Lambda = 1$ and $c_I = c_E = c$
 - key tradeoff: immediate, diminishing rewards of exploitation vs. time cost of exploration
- Functional forms and normalizations
 - $L = 1$
 - $\theta(X) = (1 + X)^\phi$
 - replace $(K, x_{o,k}, p_k)$ with continuous lognormal distribution, with mean normalized to 1
- **Group 1 – externally calibrated:** ρ, σ
- **Group 2 – offline/direct match:** $[\gamma, \lambda, \delta, \delta_F, K_R]$
 - λ : waiting time T_E is exponential with rate $\lambda + \delta_F \rightarrow T_E \sim \text{Exp}(\lambda + \delta_F)$, and $\lambda = \frac{1}{\mathbb{E}[T_E]} - \delta_F$
 - do our best to measure the wait time between clusters
- **Group 3 – internally calibrated via SMM:** $[\phi, |\mu|, c, \sigma_\ell, \bar{\Lambda}_E]$
 - we do *not* target within-patch diminishing returns

Parameter values and model fit

► Implied model properties

Parameter	Description	Value	Source / target	Data	Model
<i>External</i>					
ρ	Discount rate	0.05	Conventional value		
σ	Elasticity of substitution	4	Garcia-Macia et al. (2019)		
<i>Offline</i>					
γ	Patent arrival rate	2.28	Patents per cluster-year (patents)		
λ	Patch-finding rate	0.149	Inter-cluster wait (patents)		
δ	Product obsolescence rate	0.0331	Establishment exit (BDS)		
δ_F	Firm death rate	0.0548	Mature-firm death (BDS)		
K_R	Entry cost (labor)	22.0	Average firm size (BDS)		
<i>Online/internal (SMM)</i>					
ϕ	Patch quality slope	0.0321	Aggregate growth	0.020	0.020
$ \mu $	Depletion speed	0.0681	Mean log active years (patents)	1.24	1.31
σ_ℓ	Initial-patch dispersion	0.353	Var. log patent count (patents)	1.17	1.17
c	R&D cost	0.177	R&D intensity (Compustat)	0.043	0.043
$\bar{\Lambda}_E$	Entrant relative quality	0.768	Entrant size ratio (BDS)	0.33	0.33

Growth slowdowns: question & approach

Looking back, why did productivity growth slow?

- Alternative **explanations**:
 - A) **Deteriorating exploitation** [e.g. Bloom et al., 2020]: each patent delivers a smaller quality step ($\phi \downarrow$), patches deplete faster ($|\mu| \uparrow$)
 - B) **Deteriorating exploration** [e.g. Jones, 2009]: new patches become harder to find ($\lambda \downarrow$)
- **Identification problem**: each shift shifts standard aggregate observables similarly
- **Model**: endogenizes when to move across patches $\rightarrow$ timing has identifying content
 - A) **Deteriorating exploitation** $\rightarrow$ spells shorten, more frequent rotation
 - B) **Deteriorating exploration** $\rightarrow$ spells lengthen, slower rotation

Growth slowdowns: finding

The timing evidence points to **harder exploration**, not **worsening exploitation**.

Panel A. Duration conditional on entry

Object	Definition	1981–1989	1990–1999	2000–2009	2010–2015
Duration	First-patent date	3.16	3.97	3.68	3.56
Duration	Peak-flow date	3.40	4.17	4.01	3.86

Panel B. Entry rates

Object	Definition	1981–1989	1990–1999	2000–2009	2010–2018
First-ever entry rate	First-patent entry	1.43	1.32	1.10	0.75
Entry/re-entry rate	First entry or >5-year return	2.13	2.03	1.81	1.73

Growth accelerations: AI

Looking ahead, how might AI reshape innovation and accelerate growth?

- The theoretical case for a dramatic acceleration rests on **AI** being a **method of invention** *[e.g., Agrawal et al., 2018]*
- But does AI “merely” accelerate search within domains researchers already work in, or can its breadth of knowledge open genuinely new territory? Does it matter?
- Existing studies are inconclusive at best *[e.g. Boiko et al., 2023; Hill-Stein, 2026]*
- Model lens: **accelerated exploitation** ($\gamma \uparrow$) vs. **easier exploration** ($\lambda \uparrow$)
 - **both channels raise g** – AI-as-exploitation need not be a lesser force in that sense
 - but **exploitation-AI leaves growth hinging on humans continuing to find new patches**
 - and it would narrow innovation still further to familiar territory

Modes of growth: the vintage composition of innovation

What fraction of growth derives from firms exploiting new rather than old patches?

- **Model-implied statistic:** $S_{\text{new}}(h; w)$ is the w -weighted share of patents over $(\tau, \tau + h]$ originating in patches first entered in or after τ
- **Vintage composition of growth:** weighting patents by quality improvement $\Delta \tilde{q}$ yields the quality-improvement component of growth, so we can decompose it

$$\Phi \equiv \sum_{\text{patents}} \Delta \tilde{q}, \quad (\sigma - 1) g_{\text{pat}} \equiv \frac{1}{\mathcal{N}} \int \gamma [\theta(x)^{\sigma-1} - 1] \Phi_z(x) dx = \frac{\Phi}{\tilde{Q}}$$

where $\tilde{Q} \equiv Q^{\sigma-1} = \int_0^{\mathcal{N}} \tilde{q}_j dj$ is the aggregate quality stock

What fraction of growth derives from firms exploiting new rather than old patches?

Patent weight	Model		Data	
	$h = 10$	$h = 20$	$h = 10$	$h = 20$
<i>A. Share of patenting from new clusters</i>				
Count	0.64	0.81	0.37	0.51
Quality increment	0.44	0.65	–	–
Private value	0.45	0.66	0.40	0.49

Firms continually entering new territory sustains growth.

How does the composition of innovation differ across economies?

- **Draghi report** (2024): US innovation in newest tech, European firms “specialised in mature technologies where the potential for breakthroughs is limited.”
- Model counterpart: share of growth originating in patches firms have newly entered
 - caveat: new-to-the-firm vs. new-to-the-economy
- **Counterfactual:** take model calibrated to the U.S., lower both λ and $|\mu|$ so that g is fixed but $S_{\text{qual}}(10)$ falls by 10ppts
- In the “**higher-exploitation**” economy (Germany?)
 - by construction, **growth is not lower**
 - firms typically stay active in any one patch for longer
 - **growth more due to incremental quality improvements**, less due to new varieties
- Varieties of Capitalism [Hall-Soskice, 2001]

Conclusion

"Foraging"?

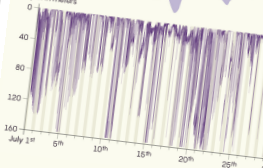
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Random Rise and Fall of a Basking Shark

When researchers monitored the vertical movements of basking sharks roaming the ocean for plankton, they found that the distribution of depth changes conformed to a classic Lévy walk.

One month of swimming depths for a basking shark tagged in 2001

Depth in meters



SW Infographics for Quanta Magazine; source: The Marine Biological Association

Sharks

Anim. Behav., 1974, 22, 953–964

HUNTING BY EXPECTATION OR OPTIMAL FORAGING? A STUDY OF PATCH USE BY CHICKADEES

By JOHN R. KREBS,* JOHN C. RYAN & ERIC L. CHARNOV†

Institute of Animal Resource Ecology, University of British Columbia, Vancouver 8

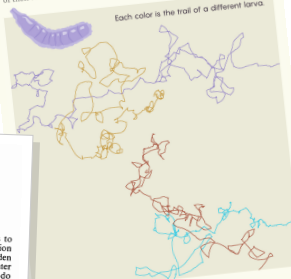
Abstract. When a predator exploits an environment in which food is patchily distributed, it has to continually make the decision of how long to stay in a patch. In this paper we examine this question using captive black-capped chickadees foraging in a large aviary for small pieces of mealworm hidden in artificial pine cones. The results of our experiments show the following: (a) when the birds encounter a long sequence of patches (groups of pine cones) each containing the same number of prey, they do not learn to expect a fixed number of prey per patch (Gibb's hypothesis of hunting by expectation); (b) similarly, the birds do not learn to spend constant time on each patch (although the data on this are patchy); (c) the birds have a constant giving-up time (interval between last catch and leaving the patch) for all types of patches within an environment, and the giving-up time is inversely related to the average capture rate for the environment. These two findings are consistent with an optimal foraging model of patch use developed by E. L. Charnov. Thus our results are more consistent with the predictions of the optimal foraging than with the hypothesis of hunting by expectation.

Krebs et al. (1974)

Larvae and Lévy Walks

Genetically engineered fruit fly larvae crawl in response to intrinsic neural signals even in the absence of stimuli from the brain or senses. The paths of their movements show Lévy walks.

Each color is the trail of a different larva.



SW Infographics for Quanta Magazine; source: Jeroen Bares

Larvae

An empirically grounded framework where growth arises from firms foraging in idea space.

- ① Systematic evidence for diminishing returns to local exploitation
- ② Historical productivity slowdown more likely due to harder exploration
- ③ Firms continually entering new territory sustains growth

Fruitful directions for future research: spillovers, firm heterogeneity, IO of patches, ...

Thank You!

Extra Slides

What's new? Contributions to the literature.

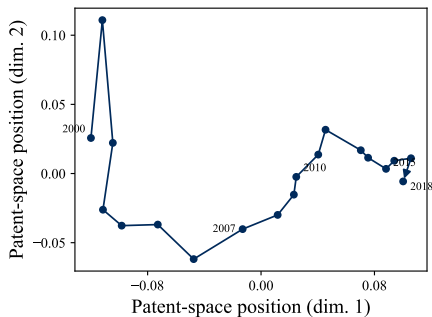
- **Modern endogenous growth theories** [Klette-Kortum, 2004; Garcia-Macia et al., 2019; Klenow-Li, 2021; Argente et al., 2024; Akcigit-Kerr, 2018; **Acemoglu et al., 2022**; **Aghion et al., 2026**; Ribeiro, 2026]
 - ⇒ introduce depleting local innovation potential & time cost of exploration → **optimal time to move** → timing identifies causes of slowdown & measures vintage composition of growth
 - ⇒ **tractable framework with a geography of ideas** → many research directions
- **Empirical research on innovation** [e.g. Bloom et al., 2020] **especially using patent data** [Kogan et al., 2017; Kelly et al., 2021; Kalyani, 2024; Ganguli et al., 2026]
 - ⇒ apply NLP to patent text to empirically represent the idea space → **new facts about innovation dynamics, esp. diminishing returns within patches**
- **Theories of explore-exploit** [e.g., Weitzman, 1979; Callander, 2011], especially in behavioral ecology [Charnov, 1976; Bénichou et al., 2005/2006/2011]
 - ⇒ cast as a canonical optimal stopping problem → embed in a growth model, **generalizing the marginal value theorem** to outside option = equilibrium object

Sample selection and data processing

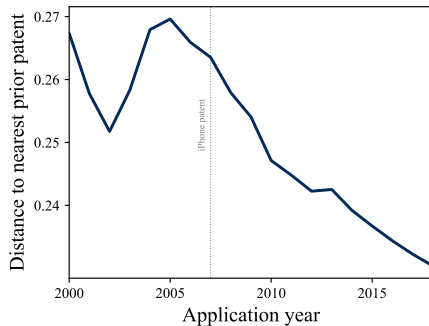
- **Patents:** granted U.S. patents from PatentsView, application years 1981–2018
 - metadata, titles and abstracts, citation links, disambiguated assignees, CPC codes
 - merge in external quality proxies: private financial value [Kogan et al., 2017], breakthrough importance [Kelly et al., 2021], creativity [Kalyani, 2024]
- **Firms:** match assignees to CRSP/Compustat for firm-year financials
- **Patches:** embed titles + abstracts, then cluster embeddings in 2 stages
 - 1 HDBSCAN within each CPC section — cluster shapes emerge from the text
 - 2 agglomerate to as many clusters as the section has CPC subclasses → 672 clusters
- **Baseline sample:** non-noise patents; public firms with ≥ 50 valid clustered patents

Apple: movements through space

Apple made large strides through the 2000s, after which movement turns local.

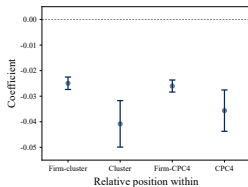


(a) Path through 2-d idea space

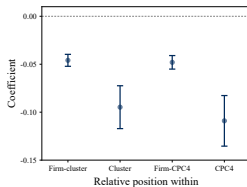


(b) Distance to nearest prior patent

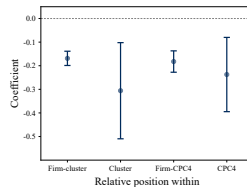
Fact 1 robustness: alternative patch definitions



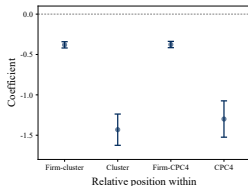
(a) Min. distance



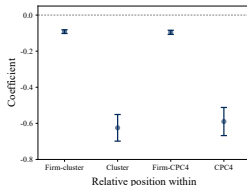
(b) Novel combinations



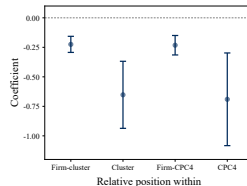
(c) Creativity



(d) Citations

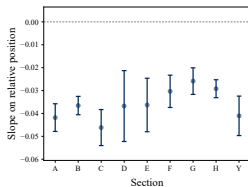


(e) Breakthrough importance

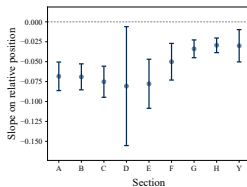


(f) Market value (KPSS)

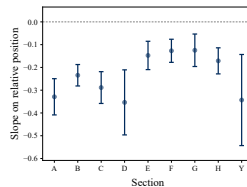
Fact 1 robustness: by CPC section

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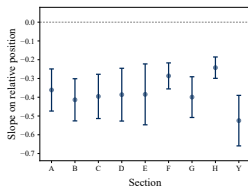
(a) Min. distance



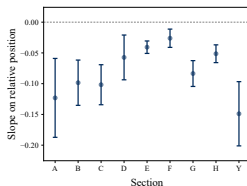
(b) Novel combinations



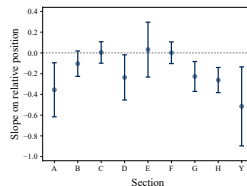
(c) Creativity



(d) Citations



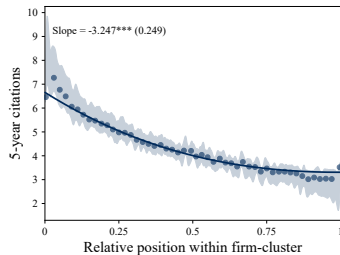
(e) Breakthrough importance



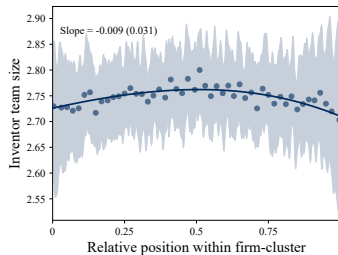
(f) Market value (KPSS)

Notes. CPC sections: A human necessities; B operations, transport; C chemistry; D textiles, paper; E fixed constructions; F mechanical engineering; G physics; H electricity; Y cross-sectional tags.

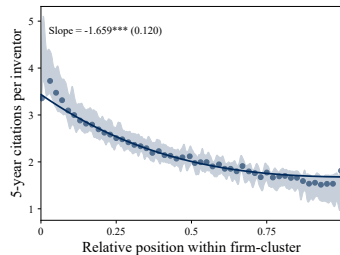
Fact 1 robustness: no declining R&D effort

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(a) Citation output



(b) Inventor team size



(c) Citations per inventor

Fact 1 robustness: selection (1/3)

Specification	Minimum distance (1)	Log citations (2)	Log KPSS value (3)	Breakthrough importance (4)	Creativity (5)	Novel combinations (6)
<i>A. Sequence endpoints</i>						
Baseline for same sequences	−0.035*** (0.001)	−0.434*** (0.027)	−0.275*** (0.043)	−0.126*** (0.008)	−0.242*** (0.024)	−0.051*** (0.004)
First patent excluded	−0.034*** (0.002)	−0.467*** (0.032)	−0.325*** (0.050)	−0.142*** (0.010)	−0.247*** (0.028)	−0.051*** (0.004)
Baseline for same sequences	−0.035*** (0.001)	−0.434*** (0.027)	−0.275*** (0.043)	−0.126*** (0.008)	−0.242*** (0.024)	−0.051*** (0.004)
Last patent excluded	−0.035*** (0.002)	−0.483*** (0.032)	−0.334*** (0.049)	−0.141*** (0.009)	−0.251*** (0.027)	−0.048*** (0.004)
Baseline for same sequences	−0.035*** (0.002)	−0.456*** (0.029)	−0.299*** (0.046)	−0.133*** (0.009)	−0.245*** (0.026)	−0.049*** (0.004)
First and last excluded	−0.034*** (0.002)	−0.504*** (0.035)	−0.367*** (0.055)	−0.153*** (0.011)	−0.254*** (0.031)	−0.048*** (0.004)

Notes. Coefficients on relative position; technology-clustered s.e.; firm-cluster and application-year FE. Retained patents keep their original positions.

Fact 1 robustness: selection (2/3)

Specification	Minimum distance (1)	Log citations (2)	Log KPSS value (3)	Breakthrough importance (4)	Creativity (5)	Novel combinations (6)
<i>B. Calendar trends</i>						
Application-year FE	−0.034*** (0.001)	−0.393*** (0.024)	−0.244*** (0.039)	−0.112*** (0.008)	−0.230*** (0.022)	−0.049*** (0.003)
Cluster × year FE	−0.031*** (0.001)	−0.238*** (0.014)	−0.150*** (0.036)	−0.029*** (0.004)	−0.138*** (0.014)	−0.041*** (0.003)
Firm × year FE	−0.039*** (0.001)	−0.475*** (0.018)	0.015* (0.009)	−0.057*** (0.005)	−0.200*** (0.017)	−0.054*** (0.004)
Both calendar FE	−0.038*** (0.001)	−0.357*** (0.014)	−0.001 (0.008)	−0.017*** (0.003)	−0.133*** (0.014)	−0.048*** (0.004)

Notes. Coefficients on relative position; technology-clustered s.e.; firm-cluster FE throughout. Same patent sample within each outcome; the calendar effects replace application-year effects.

Fact 1 robustness: selection (3/3)

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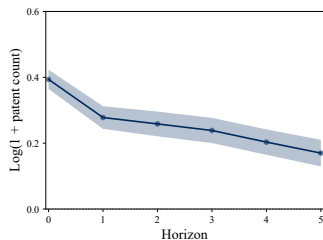
Specification	Minimum distance (1)	Log citations (2)	Log KPSS value (3)	Breakthrough importance (4)	Creativity (5)	Novel combinations (6)
<i>C. Early and late entrants</i>						
Early entrants (less than 5 years)	−0.034*** (0.002)	−0.634*** (0.045)	−0.469*** (0.073)	−0.151*** (0.010)	−0.259*** (0.039)	−0.040*** (0.005)
Late entrants (5 or more years)	−0.037*** (0.001)	−0.236*** (0.018)	−0.251*** (0.037)	−0.060*** (0.008)	−0.183*** (0.016)	−0.051*** (0.004)

Notes. Coefficients on relative position; technology-clustered s.e.; firm-cluster and application-year FE. Late entrants first patent in a cluster at least five years after the cluster first appears in the sample.

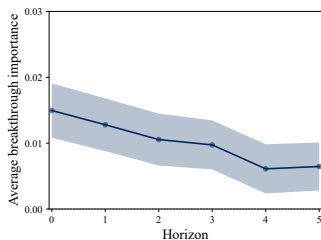
Fact 2: cluster quality and duration

	Share active years		
	(1)	(2)	(3)
Log average 5-year citations	0.0322*** (0.0013)		
Average breakthrough importance		0.0202*** (0.0057)	
Log average patent value			0.0214*** (0.0013)
Observations	28,433	25,654	28,433
R^2	0.284	0.250	0.266

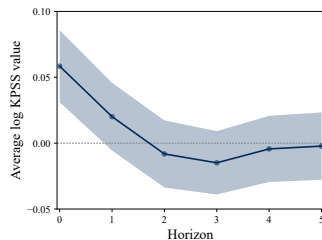
Fact 3: other local-projection outcomes

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(a) Patenting

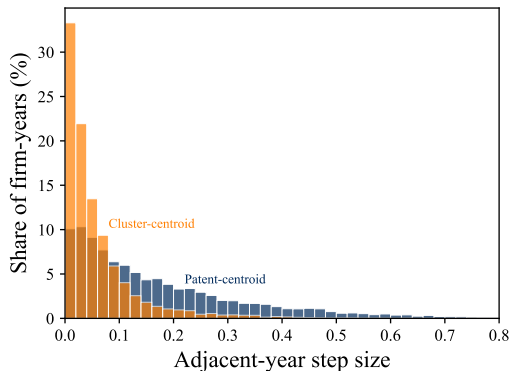


(b) Breakthrough importance



(c) Patent value

Step size distribution at the firm-year level



Notes. Step size = cosine distance between the firm's positions in adjacent years; position from patent centroids or active-cluster centroids.

Growth rate with $\delta_{\text{exit}} > 0$

- With $T \equiv \Delta/|\mu|$, spell-survival probability $P \equiv e^{-\delta_{\text{exit}} T}$, $E[T_I] = (1 - P)/\delta_{\text{exit}}$, and $P_{\text{expl}} = 1 - \delta_F E[T_I]$:

$$E[T] = \frac{1 - P}{\delta_{\text{exit}}} + \frac{P_{\text{expl}}}{\lambda + \delta_F}, \quad E[H] = \frac{\gamma\phi}{\delta_{\text{exit}}} \left[\left(x_0 + \frac{\mu}{\delta_{\text{exit}}} \right) (1 - P) + \Delta P \right]$$

- As $\delta_{\text{exit}} \rightarrow 0$: $P = 1$, $E[T] = T + 1/\lambda$, and $E[H] = \gamma\phi T \frac{x_0 + x^*}{2}$ (midpoint rule) $\rightarrow$ decomposition on main slide

Spatial overlay

- Measurement layer only – no feedback to decisions
- Exploitation = local diffusion around the patch center; patent n is observed at position $Z(T_n)$:

$$dZ(t) = \sigma_I dW_z(t).$$

- Exploration = ballistic ray at speed ν_E , direction uniform on the circle, so jump lengths are exponential:

$$f(d \mid \text{entry}) = \alpha_d e^{-\alpha_d d}, \quad \alpha_d \equiv \frac{\lambda + \delta_F}{\nu_E}.$$

BGP details (1/4): firm composition and varieties

- Notation here allows for $\sigma_x > 0$
- Given (g, L^P) , firms solve the stopping problem: HJB with diffusion term, value matching, smooth pasting $\rightarrow (\hat{W}, \hat{W}^E, x^*)$
- **Firm composition** (Little's law; ζ : flow of firms into exploitation):

$$F^I = \zeta \bar{T}^I, \quad F^E = \zeta \frac{1 - \delta_F \bar{T}^I}{\lambda + \delta_F}, \quad f_I \equiv \frac{F^I}{F} = \frac{\bar{T}^I}{\bar{T}}, \quad f_E \equiv \frac{F^E}{F} = \frac{1 - \delta_F \bar{T}^I}{(\lambda + \delta_F) \bar{T}}$$

- **Entry replaces firm death:** $E = \delta_F F$
- **Varieties:** $\dot{\mathcal{N}} = \lambda F^E + E - (\delta + \delta_F) \mathcal{N} = 0 \rightarrow$ product-to-firm ratio

$$\chi_{\mathcal{N}} \equiv \frac{\mathcal{N}}{F} = \frac{\lambda f_E + \delta_F}{\delta + \delta_F}$$

BGP details (2/4): stationary quality weights

- Average-quality weight of product j : $z_j \equiv (q_j/\bar{Q})^{\sigma-1}$; cross-sectional mean = 1 on a BGP
- Two within-spell objects (Feynman–Kac): $\bar{\Psi}^{\text{surv}}$, expected terminal quality-weight multiplier at non-death exit; $\bar{\Xi}$, expected cumulative quality weight during exploitation

$$\bar{z}_{\text{entry}} = \frac{\delta_F \bar{T} \bar{\Lambda}_E^{\sigma-1}}{\lambda \Lambda^{\sigma-1} \bar{\Psi}^{\text{surv}}}, \quad \bar{z}_I = \bar{z}_{\text{entry}} \frac{\bar{\Xi}}{\bar{T}^I}, \quad \bar{z}_E = \bar{z}_{\text{entry}} \frac{\bar{\Psi}^{\text{surv}}}{1 - \delta_F \bar{T}^I} \frac{\lambda + \delta_F}{\lambda + \delta_F + (\sigma - 1)g}$$

- Stationarity requires: $\frac{\lambda \Lambda^{\sigma-1} \bar{\Psi}^{\text{surv}}}{\lambda + \delta_F + (\sigma - 1)g} < 1$

BGP details (3/4): aggregate growth

- **Stock form:** g keeps the average-quality-weight distribution stationary

$$\frac{\bar{z}_{\text{entry}}}{\chi_{\mathcal{N}} \bar{T}} \frac{[(\sigma - 1)g + \delta_F] \bar{\Xi} + \bar{\Psi}^{\text{surv}}}{(\sigma - 1)g + \delta + \delta_F} = 1$$

- g too low $\rightarrow$ quality weights drift up relative to $\bar{Q}$; g too high $\rightarrow$ dilution too strong

- **Flow form**

$$(\sigma - 1)g = \underbrace{\frac{1}{\mathcal{N}} \int \gamma [\theta(x)^{\sigma-1} - 1] \Phi_z(x) dx}_{\text{patenting}} + \underbrace{\lambda \Lambda^{\sigma-1} \frac{f_E}{\chi_{\mathcal{N}}} \bar{z}_E}_{\text{discovery}} \underbrace{- \delta}_{\text{obsolescence}} + \underbrace{\delta_F \left(\frac{\bar{\Lambda}_E^{\sigma-1}}{\chi_{\mathcal{N}}} - 1 \right)}_{\text{net entry}}$$

where Φ_z : stationary density of active-product quality weight

BGP details (4/4): labor market and free entry

- **R&D labor demand** is proportional to quality weight:

$$L_I = \frac{c_I}{\sigma - 1} \frac{f_I}{\chi_{\mathcal{N}}} \bar{z}_I, \quad L_E = \frac{c_E}{\sigma - 1} \frac{f_E}{\chi_{\mathcal{N}}} \bar{z}_E$$

- **Labor market clearing** ($L = L^P + L_I + L_E + K_R \delta_F F$) pins the firm mass:

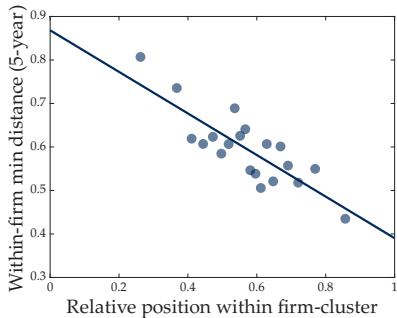
$$F = \frac{L - L^P - L_I - L_E}{K_R \delta_F}$$

- **Free entry** (pay K_R , start at $q_0 = \bar{\lambda}_E \bar{Q}$; $\bar{W}_{\text{entry}} \equiv \sum_k p_k \hat{W}(x_{0,k})$):

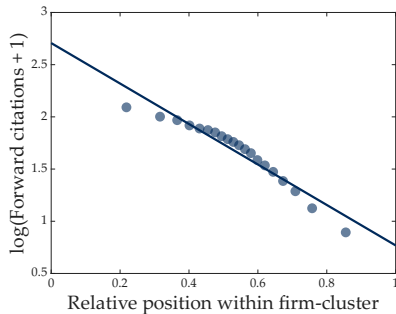
$$\chi_{\mathcal{N}} [L - L^P - L_I - L_E] = \frac{L^P \delta_F}{\sigma - 1} \bar{\lambda}_E^{\sigma-1} \bar{W}_{\text{entry}}$$

- K_R drops out, as a higher entry cost thins the firm mass but raises each entrant's value in proportion $\rightarrow$ fixes scale, but allocation unaffected

Implied model properties (1/2): within-patch diminishing returns

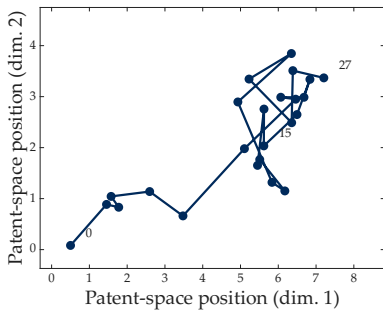


(a) Min. distance

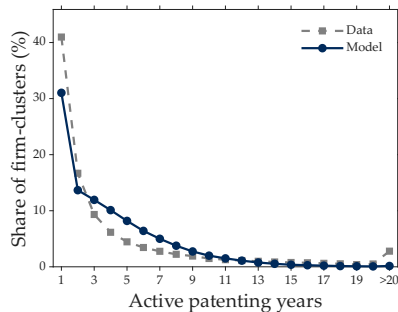


(b) Forward citations

Implied model properties (2/2): spatial movement and durations

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(a) Spatial path (one simulated firm)



(b) Active cluster durations: model vs. data

An empirical sufficient statistic

- Can we measure the vintage composition directly in the data?
- Obstacle: quality increments $\Delta\tilde{q}$ are unobserved, so S_{qual} is not directly measurable
- **Model-implied proxy:** weight patents by patent *market value* [Kogan et al., 2017]
 - model: the revaluation KPSS measure is a patent raising firm value by $\Delta V = \kappa \Delta\tilde{q} \hat{W}(X)$
 - the wedge $\hat{W}(X)$ varies little across $X \rightarrow \Delta V$ is an approx. sufficient statistic for $\Delta\tilde{q}$ up to a common factor
 - in the calibrated model, S_{val} tracks S_{qual} to within one percentage point
- **Measurement:** in the data, the value-weighted new-cluster share attributes about half of quality-improvement growth over a 20-year horizon to new clusters ← below the model share, but same order of magnitude

Modes of growth: counterfactual

	$ \mu $	λ	g	$S_{\text{qual}}(10)$	$S_{\text{qual}}(20)$	Avg. dur.	g_{pat}/g	g_{disc}/g	g/L_R	$L_E L_I$
Baseline	0.068	0.149	2.0%	0.44	0.65	3.72	134%	84%	0.397	0.020 0.030
Counterfact.	0.057	0.068	2.0%	0.34↓	0.55	4.63	161%	44%	0.318	0.023 0.040

Notes. The counterfactual holds growth fixed and lowers the new-cluster quality share at horizon 10 by 10 percentage points.